

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 3 - HACKATHON

Course Code	DSA0404	Course Title	Fundamentals of Data Science
CO Assessed	CO5 - HACKATHON	Assessment	Assessment Tool 3: Hackathon
Weightage	40% of CIA	Total Marks	100
CO4 Statement	To understand the concept of supervised and unsupervised methods in machine learning		
Student Name	Aditya Krishnan S	Reg. No.	192424069
Section/Batch	2024	Date	31 August 2026

Code of Conduct:

I, ADITYA KRISHNAN S/192424069, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Aditya Krishnan S

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Industry Problem	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyse the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Problem Statement:

Crop Selection Advisor: Using soil nutrients (N, P, K), rainfall, temperature, and humidity, build a CART model to recommend the most suitable crop for a farmer.

Code

```
DSA0404 - CO5-AT3 - Crop Advisor.py - C:/Users/Krish/OneDrive/Desktop/SIMATS/Fundamentals of Data Science/Assessment Tools/CO5/DSA0404 - CO5-AT3 - Crop Advisor.py (3.13.3)
File Edit Format Run Options Window Help

import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score

# Page configuration
st.set_page_config(
    page_title="Crop Selection Advisor",
    page_icon="🌾",
    layout="wide"
)

# Title
st.title("🌾 Crop Selection Advisor")
st.write(
    "CART-based crop recommendation using soil nutrients "
    "and environmental conditions."
)

# Load dataset
DATASET = "DSA0404 - CO5-AT3 - Dataset.csv"

try:
    df = pd.read_csv(DATASET)
except FileNotFoundError:
    st.error(f"Dataset not found: {DATASET}")
    st.stop()

# Features and target
features = [
    "N",
    "P",
    "K",
    "rainfall",
    "temperature",
    "humidity"
]

target = "crop"
```

```
DSA0404 - CO5-AT3 - Crop Advisor.py - C:/Users/Krish/OneDrive/Desktop/SIMATS/Fundamentals of Data Science/Assessment Tools/CO5/DSA0404 - CO5-AT3 - Crop Advisor.py (3.13.3)
File Edit Format Run Options Window Help

X = df[features]
y = df[target]

# Train CART model
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

model = DecisionTreeClassifier(
    criterion="gini",
    random_state=42
)

model.fit(X_train, y_train)

# Model accuracy
accuracy = accuracy_score(y_test, model.predict(X_test))

# Sidebar inputs
st.sidebar.header("🌱 Farm Conditions")

N = st.sidebar.number_input(
    "Nitrogen (N)",
    min_value=0.0,
    max_value=200.0,
    value=50.0
)

P = st.sidebar.number_input(
    "Phosphorus (P)",
    min_value=0.0,
    max_value=150.0,
    value=40.0
)
```

```
K = st.sidebar.number_input(
    "Potassium (K)",
    min_value=0.0,
    max_value=150.0,
    value=40.0
)

rainfall = st.sidebar.number_input(
    "Rainfall (mm)",
    min_value=0.0,
    max_value=300.0,
    value=100.0
)

temperature = st.sidebar.number_input(
    "Temperature (°C)",
    min_value=0.0,
    max_value=50.0,
    value=25.0
)

humidity = st.sidebar.number_input(
    "Humidity (%)",
    min_value=0.0,
    max_value=100.0,
    value=70.0
)

# Prediction
input_data = pd.DataFrame([[
    N, P, K, rainfall, temperature, humidity
]], columns=features)

prediction = model.predict(input_data)[0]
probabilities = model.predict_proba(input_data)[0]

# Main prediction section
st.subheader("🌾 Recommended Crop")

st.success(f"### {prediction}")
```

```
confidence = probabilities.max() * 100
st.write(f"Prediction confidence: {(confidence:.2f)}%")

# Model information
col1, col2, col3 = st.columns(3)

with col1:
    st.metric("Dataset Records", len(df))

with col2:
    st.metric("Crop Classes", y.nunique())

with col3:
    st.metric("Test Accuracy", f"{accuracy * 100:.2f}%")

# Prediction probabilities
st.subheader("📊 Crop Prediction Probabilities")

probability_df = pd.DataFrame({
    "Crop": model.classes_,
    "Probability": probabilities
})

probability_df = probability_df.sort_values(
    "Probability",
    ascending=False
).head(5)

st.bar_chart(
    probability_df.set_index("Crop")
)

# Feature importance
st.subheader("📈 Feature Importance")

importance_df = pd.DataFrame({
    "Feature": features,
    "Importance": model.feature_importances_
}).sort_values(
    "Importance",
    ascending=False
)
```

```
fig1, ax1 = plt.subplots(figsize=(8, 4))
ax1.bar(
    importance_df["Feature"],
    importance_df["Importance"]
)
ax1.set_xlabel("Features")
ax1.set_ylabel("Importance")
ax1.set_title("CART Feature Importance")
plt.xticks(rotation=30)
plt.tight_layout()

st.pyplot(fig1)

# Decision tree
# Decision tree
st.subheader("🌱 CART Decision Tree")

fig2, ax2 = plt.subplots(figsize=(20, 10))

plot_tree(
    model,
    feature_names=features,
    class_names=model.classes_,
    filled=True,
    rounded=True,
    max_depth=3,
    fontsize=9,
    ax=ax2
)

ax2.set_title("CART Decision Tree (Top 3 Levels)", fontsize=14)
plt.tight_layout()

st.pyplot(fig2)
plt.close(fig2)

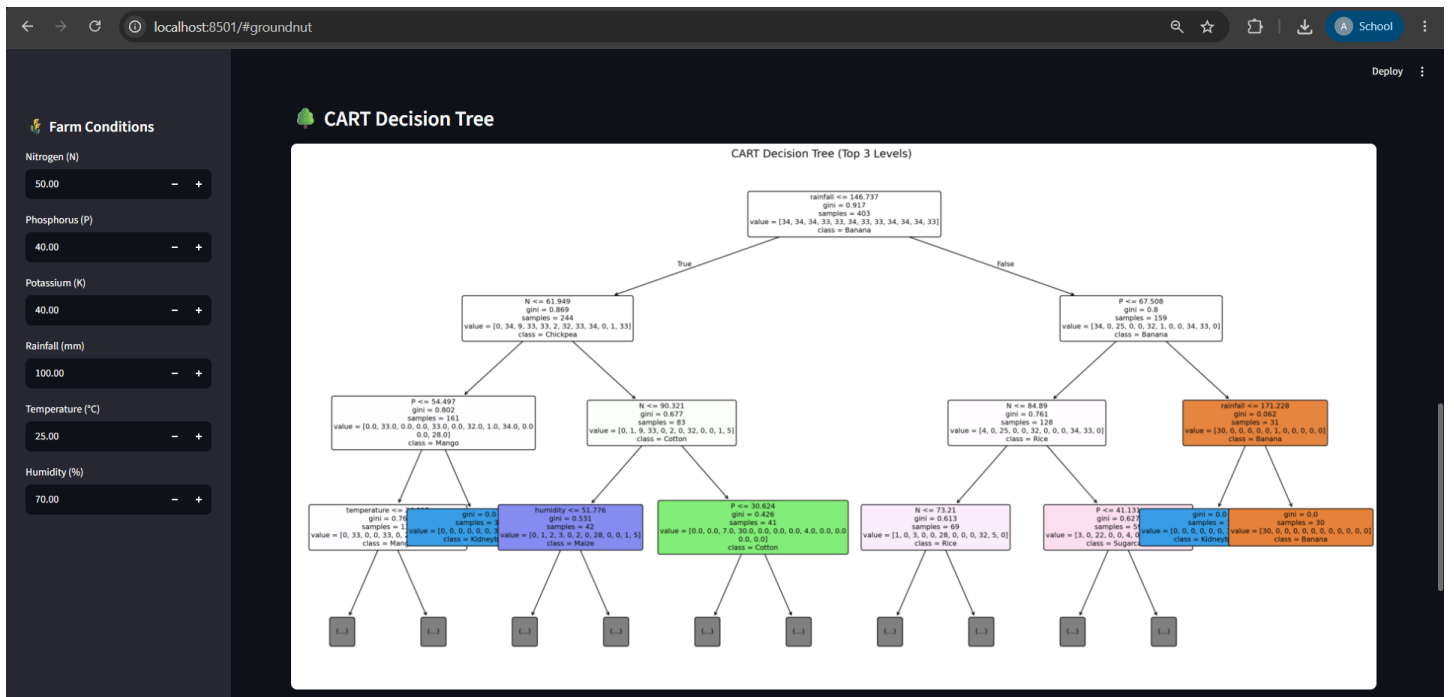
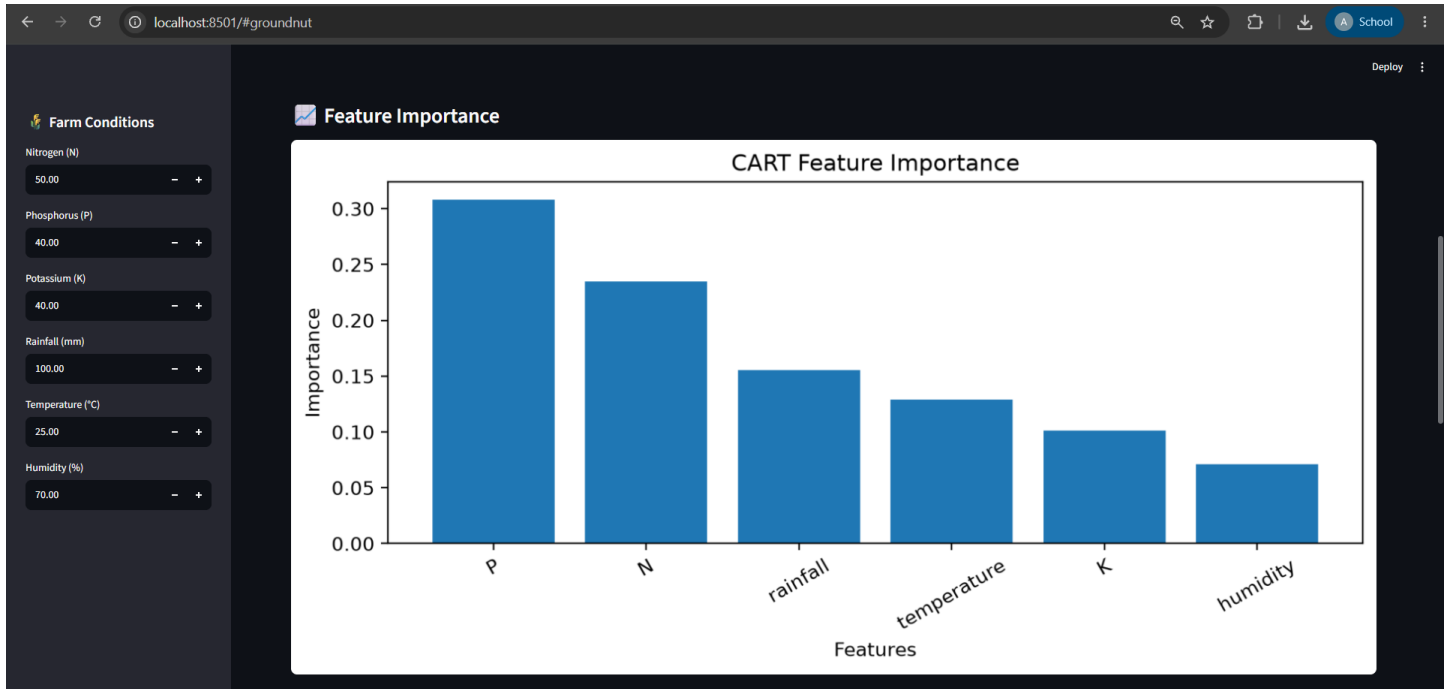
# Dataset preview
with st.expander("View Dataset"):
    st.dataframe(df)

st.caption(
    "Model: CART Decision Tree | Criterion: Gini Impurity"
)
```

Ln: 207 Col: 1

Output:





GeoTagged Image:



Result:

The CART-based Crop Selection Advisor was successfully implemented using soil nutrients (N, P, K), rainfall, temperature, and humidity as input features. The model successfully recommends a suitable crop based on the given farm conditions. The system achieved a test accuracy of 71.29% on 504 dataset records covering 12 crop classes. Feature importance and the CART decision tree were also visualized to support interpretation of the model.
